

43. Ideal Differentiation and the Different

Journal f. d. reine u. angew. Math. 188 (1950), pp. 1–21

Introduction

The principal theorem of ramification theory for algebraic number fields says, as is well known, that the different, the ramification ideal, is divisible at least by the $(e - 1)$ -st power of a prime ideal which occurs to the e -th power in the prime number p ; precisely by the $(e - 1)$ -st power when e is not divisible by p , and by a higher power when e is divisible by p .

This is analogous to the fact that the derivative $f'(x)$ of a polynomial $f(x)$ is divisible at least by the $(e - 1)$ -st power of a linear factor which occurs in $f(x)$ to the e -th power; precisely by that power when e is not divisible by the characteristic of the coefficient domain, and by a higher power when e is divisible by the characteristic.

I shall show that this is more than a formal analogy. The different can be regarded as the differential quotient of a defining ideal of the number field K in an associated integral polynomial domain in $x_1, \dots, x_n$, the differential quotient being taken at the point $x = \omega$, that is, at $x_1 = \omega_1, \dots, x_n = \omega_n$, where $\omega_1, \dots, \omega_n$ is a module basis of the system of algebraic integers of K , the principal order $\mathfrak{o}$. The defining ideal $\mathfrak{M}$ is the totality of the relations among the ω_i , namely all integral polynomials $f(x)$ such that $f(\omega) = 0$. The differential quotient $\mathfrak{M}'[x \rightarrow \omega]$ is defined as a difference quotient at $x = \omega$.

Indeed, because $f(\omega) = 0$, regard $\mathfrak{M}$ as consisting of all differences $f(x) - f(\omega)$. Form the difference ideal

$$\mathfrak{B} = (x_1 - \omega_1, \dots, x_n - \omega_n)$$

and the difference quotient

$$\mathfrak{A} = \mathfrak{M} : \mathfrak{B},$$

where the quotient is the usual ideal quotient, taken in the polynomial domain in the x_i with coefficients in the integers of K , hence in $\mathfrak{o}$. The different $\mathfrak{D}$ is then defined by

$$\mathfrak{D} = \mathfrak{A}[x \rightarrow \omega].$$

The extension of the coefficient domain is necessary in order for the difference ideal and the difference quotient to be defined. Thus this is the direct generalization of the differential quotient of a polynomial in one variable,

$$f'(\xi) = \left. \frac{f(x) - f(\xi)}{x - \xi} \right|_{x=\xi},$$

where here as well the polynomial domain must be enlarged by adjoining ξ so that the differences are defined. In fact the ideal differential quotient becomes the ordinary polynomial differential quotient as soon as the basis of the ω_i consists of the powers of a single element.

The definition just given therefore attaches directly to familiar facts. It introduces, however, non-invariant intermediate objects, since the ideal $\mathfrak{M}$ depends on the chosen basis. An equivalent, completely invariant definition is obtained by replacing the integral polynomial domain by a ring $\mathfrak{D}$ isomorphic to the principal order $\mathfrak{o}$; the coefficient domain of $\mathfrak{D}$ is then extended by $\mathfrak{o}$ to form $\mathfrak{D}_{\mathfrak{o}}$. The difference ideal is generated by all differences $x - \xi$, where x and ξ correspond under the isomorphism from $\mathfrak{D}$ to $\mathfrak{o}$; the difference quotient is the quotient of the zero ideal by this ideal; and the different is obtained by replacing each x in the difference quotient by the corresponding ξ .

This invariant definition is placed first below. It defines at once the different of an arbitrary commutative ring relative to a subring, subject only to hypotheses ensuring that coefficient extension is possible. These hypotheses are automatically satisfied, for example, when a module basis exists, possibly after passage to suitable quotient rings. Thus one obtains, in particular, the different of an arbitrary order in a number field, the different of a residue class ring of an order modulo a prime-power, relative differentials, and differentials for algebraic functions of several indeterminates.

The agreement of this differential definition with the usual definition follows from a structural analysis of the Galois extension ring attached to the number field by direct product formation. This analysis is based on a direct-sum decomposition. In the special case of a residue class ring modulo a polynomial in one indeterminate, it becomes an analysis of the Lagrange interpolation formula.

More precisely, introduce a ring $\mathfrak{K}$ isomorphic to the number field K and containing $\mathfrak{o}$, and extend its coefficient field, the rational field, by the Galois closure F of K . The extended system $\mathfrak{K}_F$ is a direct sum of n simple components corresponding to the n isomorphisms of K . These components are extensions of the difference quotient and of its conjugates; at the same time the zero ideal of $\mathfrak{K}_F$ is the intersection of the n ideals obtained from the difference ideal and its conjugates. The simple components are of the form $Fe^{(i)}$, where the $e^{(i)}$ are the components of the identity. Since $e^{(i)}$ goes over to the identity under $x = \xi$, the difference quotient is $\mathfrak{d}e^{(i)}$, where $\mathfrak{d}$ denotes the different of $\mathfrak{o}$. Thus $\mathfrak{d}$ consists of precisely those elements of the field which, when multiplied by $e^{(i)}$, lie in the direct product of the order with itself.

To recover the definition by the complementary module, observe that $e^{(i)}$ is a linear form in the basis $x_1, \dots, x_n$ corresponding to the ω_i . The coefficients of the x_i in the $e^{(i)}$ form a basis of the complementary module of $\mathfrak{o}$ and of its conjugates. Since the difference quotient has coefficients in $\mathfrak{o}$, the different is the quotient of $\mathfrak{o}$ by its complementary module. This is Dedekind's definition. The construction is not restricted to the principal order; it characterizes the different of every order as the quotient of the order by its complementary module.

In the principal-order case, the agreement with the definition by a fundamental equation follows from the fact that the ideal differential quotient becomes the polynomial differential quotient when the basis consists of powers of one element. The conceptual relation between this definition and Dedekind's is thereby obtained.

With a fundamental equation one obtains directly the principal ramification theorem mentioned at the start. For the exact determination of exponents one considers the different $\mathfrak{d}[p^t]$ of the residue class ring of the principal order modulo p^t , with t sufficiently large; $t = n$ is enough, and for quadratic fields is also necessary. The crucial point is that the different $\mathfrak{d}$, reduced modulo p^t , passes exactly into $\mathfrak{d}[p^t]$. One may then restrict to the residue class ring modulo p^t . This ring is isomorphic to a residue class ring modulo a polynomial in one indeterminate with coefficients modulo p^t ; differentiating this polynomial gives, as O. Ore showed by another route, the exact survey of possible exponents by means of the supplement numbers.

Finally, the structural analysis also gives the known theorem that the different of an overfield of K is the product of the relative different and the different of K . This is a consequence of the corresponding multiplication property for the identity components of the direct-sum decomposition, and therefore for complementary modules. All the preceding facts hold in the same manner for relative differentials after passage to quotient rings, and for algebraic function fields with coefficient field of characteristic zero. In characteristic p , and for integral algebraic functions, the structural properties remain, but the ramification theory changes because inseparable extensions can occur.

§1. Extension of the Coefficient Domain of a Ring, or Direct Product Formation. Defining Ideals

In this section general rings are considered; multiplication is not assumed commutative. Unless the contrary is explicitly stated, however, the existence of an identity element is assumed.

1. Definition of the direct product. A ring $\mathfrak{D} \times_{\mathfrak{h}} \mathfrak{o}$, or $\mathfrak{D}_{\mathfrak{o}}$, is called the direct product of $\mathfrak{D}$ and $\mathfrak{o}$ over $\mathfrak{h}$, or the ring obtained from $\mathfrak{D}$ by extension of the coefficient domain $\mathfrak{h}$ to $\mathfrak{o}$, if the following conditions hold.

1. It contains $\mathfrak{D}$ and $\mathfrak{o}$ as subrings and is generated as their product.
2. The intersection of $\mathfrak{D}$ and $\mathfrak{o}$ is $\mathfrak{h}$; the identity lies in $\mathfrak{h}$.
3. The elements of $\mathfrak{D}$ commute elementwise with the elements of $\mathfrak{o}$. Hence every element is a finite bilinear combination

$$x_1 y_1 + \dots + x_m y_m, \quad x_i \in \mathfrak{D}, \quad y_i \in \mathfrak{o}.$$

4. Equality is only that forced formally by the relations in the two factors. Thus a relation

$$\sum_i x_i y_i = 0$$

must be reducible, by adding formally vanishing expressions $(yh)\delta - y(h\delta)$, to relations inside $\mathfrak{D}$ and inside $\mathfrak{o}$ separately.

These requirements determine equality, addition, and multiplication uniquely from the corresponding laws in the factors. Isomorphic data give isomorphic direct products. Conversely, any ring which is a product of $\mathfrak{D}$ and $\mathfrak{o}$ with these two factors commuting elementwise is a homomorphic image of the direct product.

2. Criterion for existence. The direct product need not exist. For example, take $\mathfrak{h} = \mathbb{Z}$, $\mathfrak{D} = \mathfrak{h}[x]$ with relation $1 - 2x = 0$, and $\mathfrak{o} = \mathfrak{h}[y]$ with relation $1 - 2y = 0$. No common overring can have intersection exactly $\mathfrak{h}$, since in any such ring

$$0 = (1 - 2x)y - x(1 - 2y) = y - x.$$

The necessary and sufficient condition is that there be at least one ring R containing $\mathfrak{D}$ and $\mathfrak{o}$, in which their intersection is $\mathfrak{h}$ and in which they commute elementwise. If the direct product exists, it itself is such a ring. Conversely, one constructs the direct product from formal bilinear combinations with the equality described above; the embedding ring R supplies the required intersection property.

3. Defining ideals. A system S of elements z of $\mathfrak{D}$ is a generating system of $\mathfrak{D}$ over $\mathfrak{h}$ if $\mathfrak{D} = \mathfrak{h}[S]$. If $\mathfrak{h}$ lies in the center of $\mathfrak{D}$, then $\mathfrak{D}$ is a homomorphic image of the noncommutative polynomial domain $\mathfrak{h}\{Z\}$, where the indeterminates Z correspond to the elements of S . Thus

$$\mathfrak{D} \simeq \mathfrak{h}\{Z\}/\mathfrak{M},$$

where $\mathfrak{M}$ is the two-sided ideal of all polynomials $F(Z)$ for which $F(z) = 0$ in $\mathfrak{D}$. This is the defining ideal. If there is a single generator z and $\mathfrak{M}$ is principal, a generator $G(Z)$ with $G(z) = 0$ is a defining equation.

4. The defining ideal of a direct product. Let $\mathfrak{D} \simeq \mathfrak{h}\{Z\}/\mathfrak{M}$. In $\mathfrak{o}\{Z\}$ let $\mathfrak{M}_\mathfrak{o}$ be the ideal generated by $\mathfrak{M}$ after extension of coefficients. If the direct product $\mathfrak{D}_\mathfrak{o}$ exists, then

$$\mathfrak{D}_\mathfrak{o} \simeq \mathfrak{o}\{Z\}/\mathfrak{M}_\mathfrak{o}.$$

Thus the defining relations in the direct product are exactly those obtained by extending the coefficients in the defining ideal.

§2. Rings with Independent Module Bases. Construction of the Direct Product

1. Construction. A system T of elements t of $\mathfrak{D}$ is an $\mathfrak{h}$ -module basis if every element of $\mathfrak{D}$ is a finite linear combination

$$h_1 t_1 + \cdots + h_m t_m, \quad h_i \in \mathfrak{h}.$$

It is independent if $\sum h_i t_i = 0$ implies all $h_i = 0$. If $\mathfrak{D}$ has an independent $\mathfrak{h}$ -module basis containing the identity, then the direct product $\mathfrak{D}_\mathfrak{o}$ exists for every extension ring $\mathfrak{o}$ of $\mathfrak{h}$ whose intersection with $\mathfrak{D}$ is $\mathfrak{h}$ and whose elements commute with those of $\mathfrak{D}$. It is the ring of all expressions

$$y_1 t_1 + \cdots + y_m t_m, \quad y_i \in \mathfrak{o},$$

with equality by coefficients in the basis T .

2. Extension and contraction of modules. For an $\mathfrak{h}$ -module $\mathfrak{B} \subset \mathfrak{D}$, let $\mathfrak{B}_\mathfrak{o}$ be the $\mathfrak{o}$ -module generated by it in $\mathfrak{D}_\mathfrak{o}$. For an $\mathfrak{o}$ -module $\mathfrak{C} \subset \mathfrak{D}_\mathfrak{o}$, its contraction is $[\mathfrak{C}, \mathfrak{D}] = \mathfrak{C} \cap \mathfrak{D}$. If $\mathfrak{B}$ has an independent $\mathfrak{h}$ -module basis extendable to such a basis of $\mathfrak{D}$, then

$$\mathfrak{B} = [\mathfrak{B}_\mathfrak{o}, \mathfrak{D}].$$

The proof is simply comparison of coefficients in the extended basis.

3. A sufficient criterion. If there exists an extension ring $\mathfrak{f}$ of $\mathfrak{h}$ such that the direct products $\mathfrak{D}_\mathfrak{f}$ and $\mathfrak{o}_\mathfrak{f}$ exist over $\mathfrak{h}$, and such that $\mathfrak{D}_\mathfrak{f} \times_\mathfrak{f} \mathfrak{o}_\mathfrak{f}$ exists over $\mathfrak{f}$, then $\mathfrak{D} \times_\mathfrak{h} \mathfrak{o}$ exists. Indeed the latter product gives an embedding ring and

$$[\mathfrak{D}, \mathfrak{o}] \subseteq [\mathfrak{D}_\mathfrak{f}, \mathfrak{o}_\mathfrak{f}] = \mathfrak{f}, \quad [\mathfrak{D}, \mathfrak{f}] = \mathfrak{h}.$$

§3. The Different of a Ring over a Subring. Differential Quotient of a Defining Ideal

From now on all rings are commutative. The rings $\mathfrak{D}$ and $\mathfrak{o}$ are assumed to be isomorphic, equivalent extensions of $\mathfrak{h}$, and the direct product $\mathfrak{D}_\mathfrak{o}$ is assumed to exist.

1. Definition of the different. Let x run through the elements of $\mathfrak{D}$, and let ξ be the corresponding element of $\mathfrak{o}$. The ideal

$$\mathfrak{B} = \{\dots, x - \xi, \dots\}$$

generated by all differences is the difference ideal. The difference quotient is

$$\mathfrak{A} = (0) : \mathfrak{B}.$$

The different of $\mathfrak{o}$ over $\mathfrak{h}$ is

$$\mathfrak{d} = \mathfrak{A}[x \rightarrow \xi],$$

and the different of $\mathfrak{D}$ is $\mathfrak{D} = \mathfrak{A}[\xi \rightarrow x]$.

2. Differential quotient of a defining ideal. Let $\mathfrak{M}$ be the defining ideal of $\mathfrak{D}$ over $\mathfrak{h}$ with respect to generators z . In $\mathfrak{o}[Z]$ put

$$\mathfrak{B}_Z = \{\dots, Z - \xi, \dots\}, \quad \mathfrak{C} = \mathfrak{M}_0 : \mathfrak{B}_Z.$$

Equivalently,

$$\mathfrak{C} = \{\dots, F(Z) - F(\xi), \dots\} : \{\dots, Z - \xi, \dots\}.$$

The differential quotient $\mathfrak{M}'$ is $\mathfrak{C}[\xi \rightarrow Z]$. It contains $\mathfrak{M}$, and

$$\mathfrak{M}'[Z \rightarrow z] = \mathfrak{D}, \quad \mathfrak{M}'[Z \rightarrow \xi] = \mathfrak{d}.$$

3. A defining equation. Suppose $\mathfrak{D}$ has a defining equation

$$G(z) = 0, \quad G(Z) = Z^n + h_1 Z^{n-1} + \dots + h_n, \quad h_i \in \mathfrak{h}.$$

Then the different is defined and is principal, with generator

$$G'(\xi) \quad \text{respectively} \quad G'(z),$$

Moreover

$$\mathfrak{M}' = (G'(Z), G(Z)).$$

The reason is that $1, z, \dots, z^{n-1}$ is an independent module basis and that

$$C(Z, \xi) = \frac{G(Z) - G(\xi)}{Z - \xi}$$

generates the difference quotient. Substitution $\xi \rightarrow Z$ gives $G'(Z)$.

§4. The Different of a Direct Sum

Assume

$$\mathfrak{D} = R_1 + \dots + R_r, \quad e = e_1 + \dots + e_r, \quad R_i R_j = 0 \ (i \neq j),$$

a direct sum of ideals, where the e_i are the components of the identity. Assume also that each R_i has an independent $\mathfrak{h}_i$ -module basis containing e_i , with $\mathfrak{h}_i = \mathfrak{h}e_i$, and that

$$[\mathfrak{h}, \mathfrak{D}_i] = 0$$

holds for the corresponding complementary summands. Then $\mathfrak{D}$ itself has an independent $\mathfrak{h}$ -module basis assembled from those of the R_i , and the different of $\mathfrak{D}$ over $\mathfrak{h}$ is the direct sum of the differentials of the R_i over the $\mathfrak{h}_i$.

If the corresponding decomposition of $\mathfrak{o}$ is

$$\mathfrak{o} = r_1 + \dots + r_r, \quad 1 = \varepsilon_1 + \dots + \varepsilon_r,$$

then

$$\mathfrak{D}_0 = \sum_{i,j} R_i \varepsilon_j.$$

The component $R_i \varepsilon_i$ carries the direct product $R_i \times r_i$, and the component of the global difference quotient on it is

$$\mathfrak{A}_{e_i \varepsilon_i} = (0) : \mathfrak{B}_{e_i \varepsilon_i}.$$

Therefore, writing $\mathfrak{D}_i$ and $\mathfrak{d}_i$ for the component differents,

$$\mathfrak{D} = \mathfrak{D}_1 + \cdots + \mathfrak{D}_r, \quad \mathfrak{d} = \mathfrak{d}_1 + \cdots + \mathfrak{d}_r.$$

The proof consists in passing between the direct-sum components by coefficient comparison in the independent bases, verifying the sharpened intersection relations

$$[\mathfrak{o}, \mathfrak{D}_i] = 0,$$

the induced isomorphisms

$$\mathfrak{o} \sim \mathfrak{o} e_i, \quad R_i \sim R_i \varepsilon_i, \quad \mathfrak{h}_i \sim \mathfrak{h}_i e_i.$$

§5. The Galois Extension Ring of a Field. Structure Theorems

1. The rings involved. Let K be a separable field extension of degree n over a ground field P . Let F be a Galois field containing the n conjugate fields

$$K^{(1)}, \dots, K^{(n)}, \quad K = K^{(1)},$$

and generated by their product. Let $\mathfrak{K}$ be an extension of P isomorphic to K , equivalent over P , and such that $[\mathfrak{K}, F] = P$. Since $\mathfrak{K}$ has a finite independent P -basis, the direct products

$$\mathfrak{K}_K, \quad \mathfrak{K}_F, \quad K_F$$

exist. The ring $\mathfrak{K}_F$ is called the Galois extension ring of $\mathfrak{K}$.

Every ideal of $\mathfrak{K}$ is the contraction of its extension in $\mathfrak{K}_K$ or $\mathfrak{K}_F$, and every ideal of $\mathfrak{K}_K$ is the contraction of its extension in $\mathfrak{K}_F$. This is the module-basis theorem of §2 over fields.

Write

$$\mathfrak{B}^{(i)} = \{\dots, x - \xi^{(i)}, \dots\}$$

for the conjugate difference ideals, and

$$\mathfrak{A}^{(i)} = (0) : \mathfrak{B}^{(i)}$$

for their difference quotients. Their extensions to $\mathfrak{K}_F$ are denoted by $\mathfrak{B}_F^{(i)}$ and $\mathfrak{A}_F^{(i)}$.

2. Direct-sum decomposition. The structure theorem is:

$$(0) = \bigcap_{i=1}^n \mathfrak{B}_F^{(i)}, \quad \mathfrak{K}_F = \mathfrak{A}_F^{(1)} + \cdots + \mathfrak{A}_F^{(n)}$$

as a direct sum. In particular,

$$\mathfrak{K}_K = \mathfrak{A} + \mathfrak{B}$$

as a direct sum, and its zero ideal is the intersection of the corresponding difference ideal with the difference quotient.

The residue class ring modulo each $\mathfrak{B}_F^{(i)}$ has rank one over F , and by separability the ideals $\mathfrak{B}_F^{(i)}$ are distinct and pairwise coprime. Hence the identity decomposes uniquely as

$$e = e^{(1)} + \cdots + e^{(n)},$$

where

$$e^{(i)} \in \bigcap_{j \neq i} \mathfrak{B}_F^{(j)}, \quad e^{(i)} \equiv 1 \pmod{\mathfrak{B}_F^{(i)}}.$$

The simple components are $Fe^{(i)}$.

3. Components as conjugates. If

$$x = \xi^{(1)}e^{(1)} + \cdots + \xi^{(n)}e^{(n)}$$

is the component representation of an element $x \in \mathfrak{K}$, then the components are conjugate. Thus $\xi^{(1)}, \dots, \xi^{(n)}$ are the n conjugate values of x under the isomorphisms of $\mathfrak{K}$ to the conjugate fields $K^{(i)}$. If B is an absolute module in $\mathfrak{K}$ and $B^{(i)}$ its components in $\mathfrak{K}_F$, then

$$B = [\mathfrak{K}, B^{(1)} + \cdots + B^{(n)}].$$

4. Complementary bases. Each P -basis $t_1, \dots, t_n$ of $\mathfrak{K}$ has a complementary basis $T_1, \dots, T_n$. It is obtained from the expansion of the identity components

$$e^{(i)} = A_1^{(i)}t_1 + \cdots + A_n^{(i)}t_n.$$

If $\alpha_1^{(i)}, \dots, \alpha_n^{(i)}$ are the conjugate bases corresponding to the t_j , then the matrices

$$(\alpha_j^{(i)})_{i,j} \quad \text{and} \quad (A_j^{(i)})_{i,j}$$

are reciprocal:

$$(\alpha_j^{(i)})(A_j^{(i)})^t = E_n.$$

The complement of the complementary basis is the original basis. Under a change $(s) = (t)P$, the complementary bases transform contragrediently:

$$(S) = P^{-1}(T).$$

If

$$c = C_1t_1 + \cdots + C_nt_n = c_1T_1 + \cdots + c_nT_n,$$

then, with Sp denoting the sum of conjugates,

$$C_i = \text{Sp}(cT_i), \quad c_i = \text{Sp}(ct_i).$$

5. A defining equation. The planned discussion of the special case of a defining equation and of Lagrange's interpolation formula was left blank in the manuscript. The preceding structure theorem gives the invariant form of the result.

§6. The Different of an Order

The structure theorems for fields are now sharpened in the integral sense. Let $\mathfrak{h}$ be an integrally closed subring of P whose quotient field is P . Let $\mathfrak{O}$ be an $\mathfrak{h}$ -order in $\mathfrak{K}$, meaning a subring containing $\mathfrak{h}$, generating $\mathfrak{K}$ as quotient field, and possessing a finite, not necessarily independent, $\mathfrak{h}$ -module basis. Let $\mathfrak{o}$ be the corresponding order in K , and $\mathfrak{o}^{(i)}$ its conjugates. Let $\mathfrak{g}$ be the ring generated by these conjugate orders in F .

The direct products $\mathfrak{O}_{\mathfrak{o}}$ and $\mathfrak{O}_{\mathfrak{g}}$ exist. Thus the different of $\mathfrak{O}$ and of $\mathfrak{o}$ over $\mathfrak{h}$ is defined. Write $\mathfrak{B}$ for the difference ideal, $\mathfrak{A} = (0) : \mathfrak{B}$ for the difference quotient, and

$$\mathfrak{D} = \mathfrak{A}[\xi \rightarrow x], \quad \mathfrak{d} = \mathfrak{A}[x \rightarrow \xi]$$

for the differents.

The structure theorem for the Galois extension ring of orders says that the difference ideal and quotient of the order extend to the corresponding ideals in the field case, and that the quotient for the order is the contraction of the extended quotient. Further,

$$\mathfrak{A} = \mathfrak{d}e^{(1)}, \quad \mathfrak{A}^{(i)} = \mathfrak{d}^{(i)}e^{(i)}.$$

Hence

$$\mathfrak{d} = [\mathfrak{o}, \mathfrak{A}^{(1)} + \cdots + \mathfrak{A}^{(n)}].$$

The different $\mathfrak{d}$ consists exactly of those elements $x \in K$ for which $xe^{(1)}$ lies in $\mathfrak{O}_{\mathfrak{o}}$. It is nonzero.

The complementary module. If $\mathfrak{O}$ has an independent $\mathfrak{h}$ -basis $t_1, \dots, t_n$, then the complementary elements $T_1, \dots, T_n$ form the basis of an $\mathfrak{h}$ -module $\mathfrak{C}$, the complementary module of $\mathfrak{O}$. Then

$$\mathfrak{d} = \mathfrak{o} : \mathfrak{C}.$$

For, if

$$e^{(1)} = A_1 t_1 + \dots + A_n t_n,$$

then $\mathfrak{d}$ is precisely the set of all elements d such that $dA_i \in \mathfrak{o}$ for all i . In the special case of a regular order with basis

$$1, z, \dots, z^{n-1},$$

one obtains

$$\mathfrak{d} = (f'(z)).$$

Equivalently, the complementary module $\mathfrak{C}$ consists of all elements c of $\mathfrak{K}$ for which the traces

$$\mathrm{Sp}(co), \quad o \in \mathfrak{O},$$

are integral, that is, lie in $\mathfrak{h}$.

Supplement. When the order has an independent $\mathfrak{h}$ -module basis, the difference quotient $\mathfrak{A}$ is the intersection of all conjugate difference ideals except the one used:

$$\mathfrak{A} = [\mathfrak{O}_{\mathfrak{o}}, \mathfrak{B}^{(2)} \cap \dots \cap \mathfrak{B}^{(n)}].$$

This follows because the difference ideal itself is then the contraction of its extension.

Fundamental equation. Suppose a fundamental equation exists, so that after adjoining indeterminates and passing to a quotient ring the order has a basis $1, u, \dots, u^{n-1}$. If

$$G(u) = 0$$

is a defining equation, then the different is obtained from the polynomial derivative $G'(u)$. Under integral closure the required multiplication of contents of coefficient ideals holds, and ramification theory can be treated by considering the fundamental equation modulo p^t .

§7. Ramification Theory of the Principal Order modulo p^t

Let a number field be given, with $\mathfrak{h}$ its ring of integers; or let one have a relative field over the quotient ring of the principal order by a prime ideal, with p the basis element of that prime ideal. After adjoining indeterminates $u_1, \dots, u_n$, both the principal order and the residue class ring modulo p^t possess the $\mathfrak{h}$ -module basis

$$1, u, \dots, u^{n-1}.$$

Thus in both cases the different is given by $G'(u)$. The different of the principal order passes modulo p^t into the different of the residue class ring modulo p^t .

If

$$p = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

then the residue class ring modulo p^t is the direct sum of the components corresponding to the residue class rings modulo $\mathfrak{p}_i^{te_i}$:

$$R = Re_1 + \dots + Re_r.$$

By §4, the different of this direct sum is the direct sum of the component differentials. Thus it is enough to treat a residue class ring modulo a single power $\mathfrak{p}^N$.

Choose ξ so that

$$\varphi(\xi) \equiv 0 \pmod{p}, \quad \varphi(\xi) \not\equiv 0 \pmod{p^2},$$

where $\varphi(x)$ is a prime function modulo p of degree f . For $t \geq 2$ replace ξ by $\xi^* = \xi e_1$, where e_1 is the first idempotent occurring in the decomposition of the residue class ring modulo p^t . Then

$$\mathfrak{p} = (p, \varphi(\xi^*)).$$

A basis modulo p^t is

$$\xi^\mu \varphi(\xi)^\nu, \quad \mu = 0, 1, \dots, f-1, \quad \nu = 0, 1, \dots, e-1.$$

Indeed, from $(\varphi(\xi), p) = \mathfrak{p}$ and $(p) = \mathfrak{p}^e$ follows

$$\varphi(\xi) = pM(\xi), \quad M(\xi) \not\equiv 0 \pmod{p},$$

so higher powers can be successively expressed through lower ones. The defining equation for this basis has the form

$$F(x) = \varphi(x)^e + pM(x),$$

where $M(x)$ is not divisible by $\varphi(x)$ modulo p . Consequently the different of the residue class ring is given by the polynomial derivative $F'(\xi)$. This reduces the determination of exponents to Ore's results, and in particular to the supplement numbers. By passing to the quotient ring the same applies to relative differentials; one adds only the statement that the different of an overfield is the product of the different of the base field and the relative different.

Received 25 October 1949.